Robonautica - Rhapsody's Robotics Challenge

About the Event

Rhapsody'25 presents IISc's Robotics Challenge, **Robonautica**, is collaboration with Artificial Intelligence and Robotics Laboratory (AIRL), IISc and MathWorks.

This competition is designed to push the boundaries of autonomous vehicle development. This competition tests programming expertise, problem-solving abilities, and collaborative teamwork of the participants in the field of autonomous navigation.

The primary goal is to foster innovation in autonomous ground vehicles, challenging participants to develop intelligent navigation software stack capable of maneuvering through indoor environments.

Challenge Overview

The competition encourages the advancement of autonomous driving technologies, sparking interest in robotics, AI-driven mobility, and real-world applications. Participants will engage in a two-level challenge, progressing from simulated navigation to real-world implementation.

Teams of four members, led by a Team Representative (Leader), will develop and implement autonomous navigation algorithms.

Competition Structure

Level 1: Autonomous Navigation in a Simulated Environment

Objective

Develop a software stack for a simulated robotic platform to navigate through a indoor virtual environment with maximum efficiency

Setup Requirements

1. A 3D simulation featuring several obstacles in the way to the Goal State.
2. Virtual checkpoints strategically placed across the environment.

Rules

1. Participants must develop algorithms for a simulated robot to traverse the environment autonomously.
2. The robot must efficiently reach all checkpoints while overcoming specific challenges.
3. The simulation ends when all checkpoints are reached, or the time limit is exceeded.

Skills Tested

1. Advanced motion planning algorithms.
2. Virtual sensor data interpretation and fusion.
3. Adaptive navigation strategies for diverse surfaces.
4. Energy efficiency optimization in route planning.

Level 2: Autonomous Navigation in a Real-World Environment

Objective

Deploy the navigation stack developed in Level 1 onto a real-world robotic platform, enabling it to navigate autonomously through the test arena.

Setup Requirements

1. A physical obstacle laden arena.
2. Fiducial markers or beacons placed strategically for navigation validation.

Rules

1. Robots must start autonomously from a designated point.
2. The objective is to navigate through the given path while detecting each marker/beacon
3. Points will be rewarded based on the number of markers reached and navigation efficiency.

Skills Tested

1. Development of adaptive control algorithms for navigation.

2. Seamless integration of simulation-tested navigation stack on a real-world robotic platform.

Timeline:

Sl. No.	Event	Timeline
1	Registrations Open	13th March
2	Round 1 (Simulation) Starts	15th March
3	Registration Closes & Round 1 Ends	17th March
4	Round 1 Results Announcement	18th March
5	Hand-on Session with Robotics Platform (Offline)	22nd March
6	Level 2 (Real-World Challenge)	23rd March